

MFE 2260 Project

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Summary of Project Objective

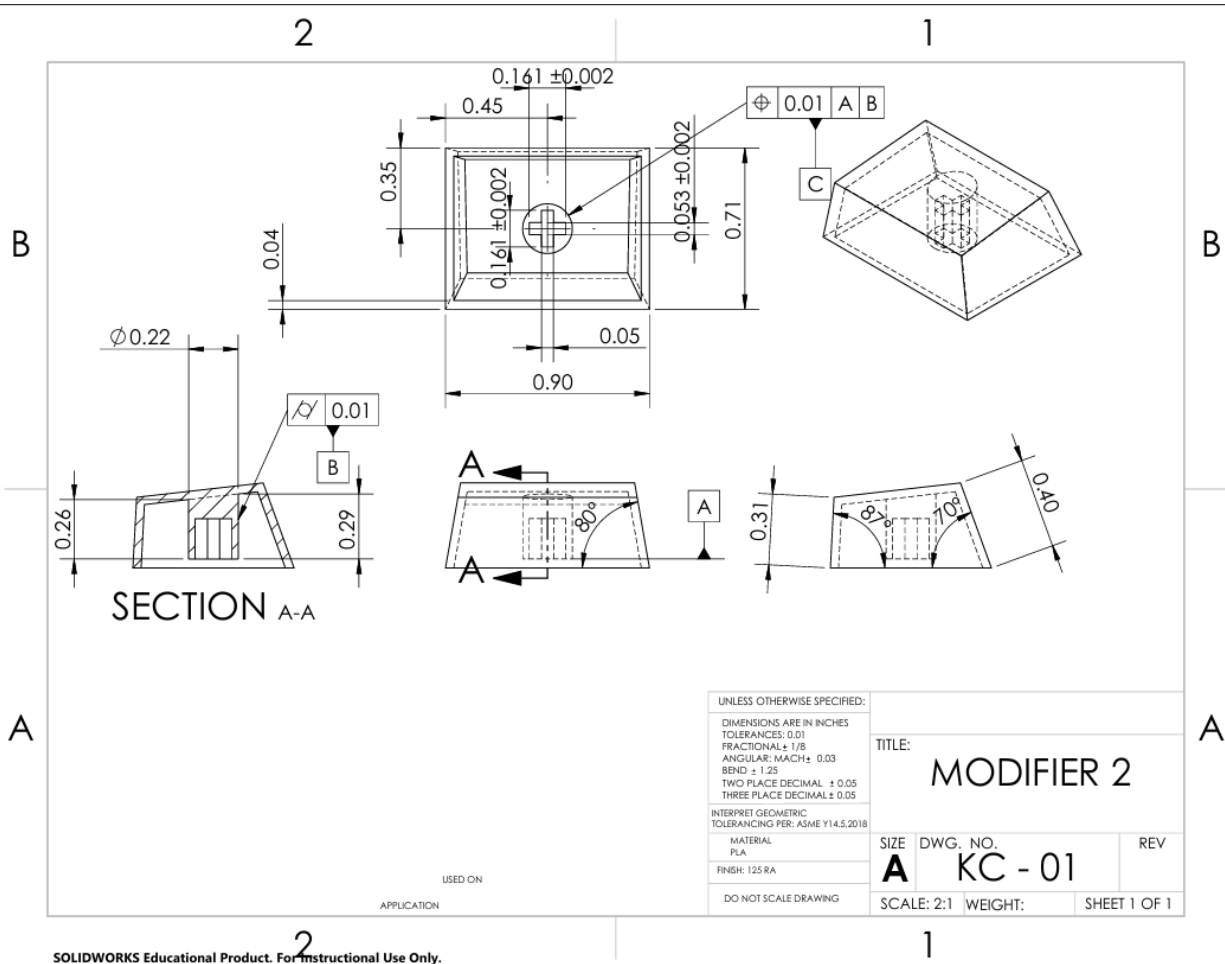
The main objective of our project was to reverse engineer a mechanical switch and stabilizer using measuring equipment to model an entire keyboard. The switch being reverse engineered is a Novelkeys cream switch. We were partially successful in this task as we were able to reseat the main components required; however, some parts proved too complicated to be recreated. Even attempts to find online recourse proved futile as we could not find consistent dimensions for the stabilizers. These parts were replaced by placeholders in order to ensure the assembly could still be built. Despite these obstacles we were able to split up the parts evenly throughout our team in order to finish them on time.

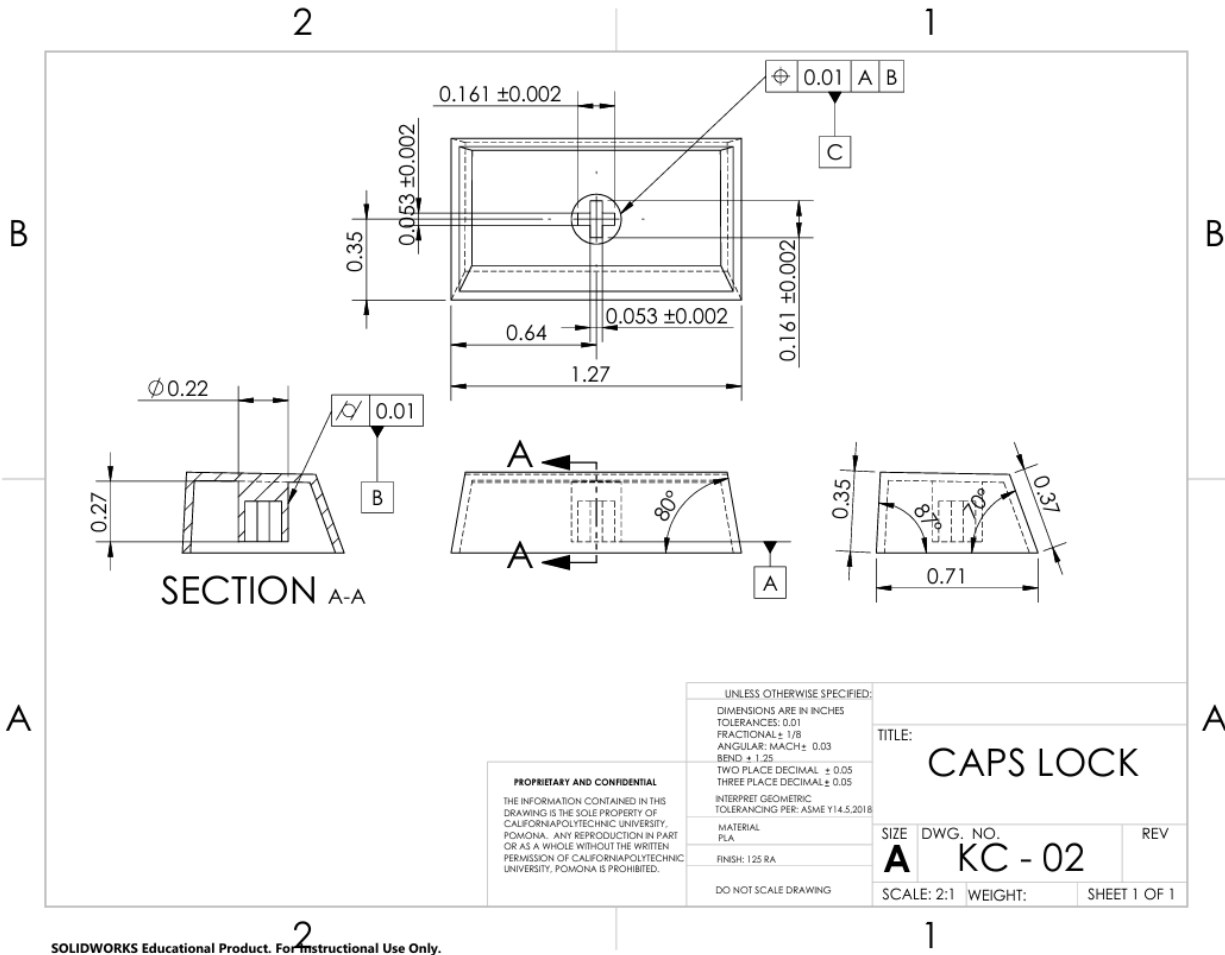
The first set of parts needed to be created were the stabilizers and keycaps. This is where we ran into problems modeling the stabilizers; however, we were able to find sufficient info on the different keycap sizes. One thing to note about keycap industry standards is that there are 9 unique templates used for each row. Our team decided on using the “Cherry” dimensioning configuration as it is one of the most widely used configurations. A lot of guessing had to be done for the key pieces but we still managed to get them to the right size.

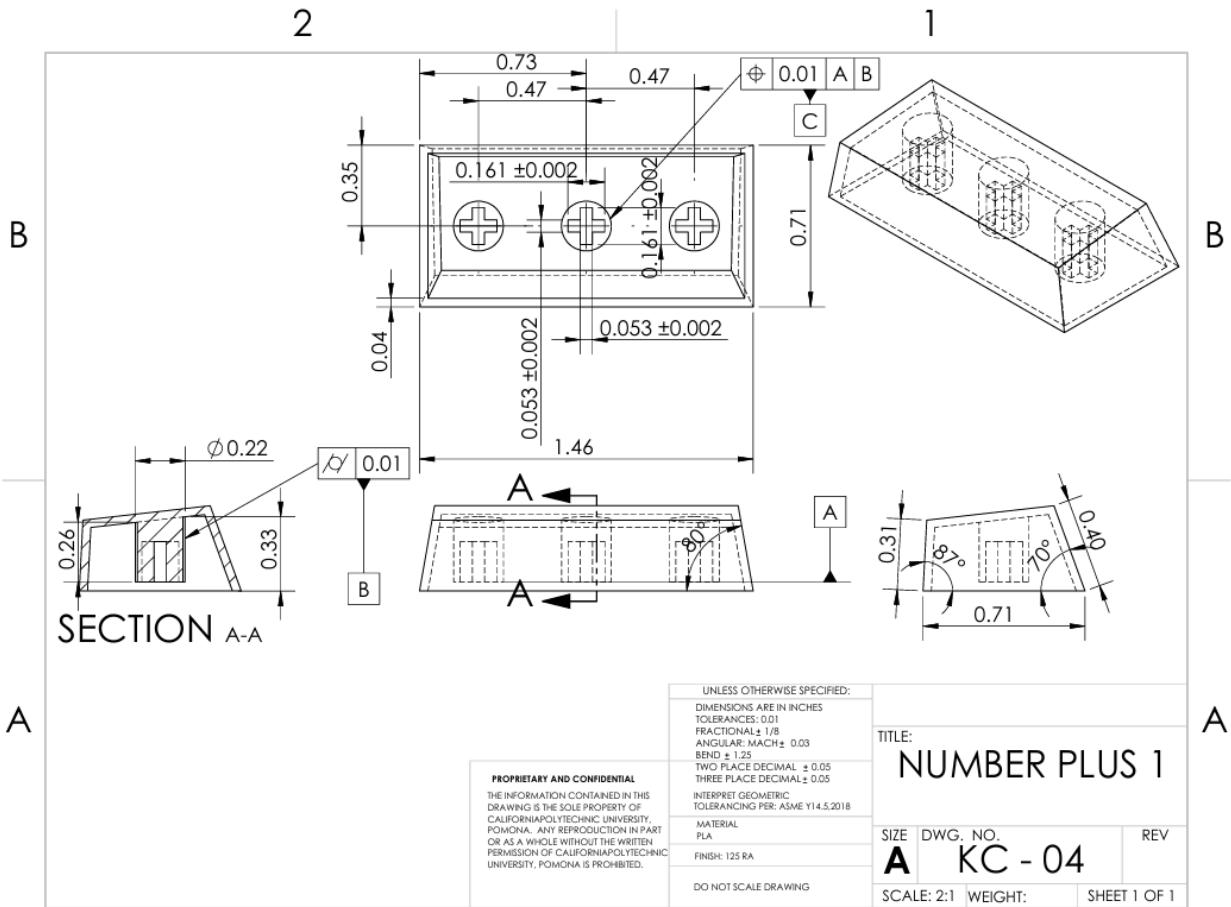
The next set of parts were the switch parts and PCB. The switch parts were easy to make but proved to be an intricate and time-consuming process. It began with closely examining the physical component, identifying its dimensions, materials, and functional parts. Turning these observations into a part required attention to detail, especially when replicating internal mechanisms like the stem and latch mechanisms for the housing. Each step revealed complexities—such as understanding tolerances, interpreting hidden internal components, and aligning every part correctly. Trial and error played a significant role, as we frequently had to revisit the design to refine inaccuracies or discover overlooked details. We also faced the

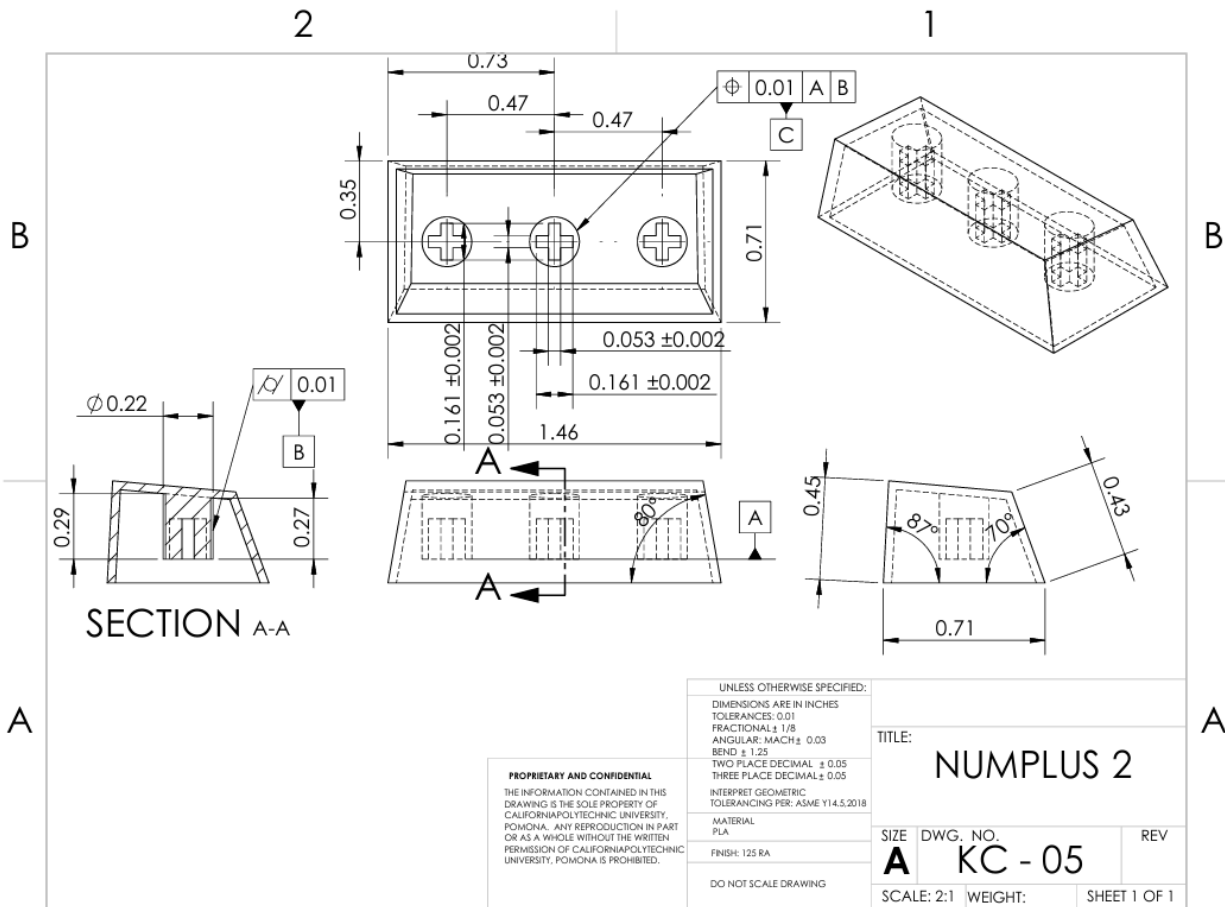
challenge of reverse engineering a PCB board from a keyboard, which proved to be an even more intricate and time-consuming process. Despite us not even attempting the electrical components this part required extreme precision and tie, as even minor inaccuracies could disrupt the entire design. We had to measure each component and trace carefully, ensuring the layout matched the original. Every step involved revisiting and refining the design to correct errors or improve accuracy, making the process feel like a test of both our patience and problem-solving skills. Approximating timeframes we would estimate that our group spent anywhere from 20 hours to measure, 15 to detail and draw, 30 to troubleshoot and revise, and 5 to assemble

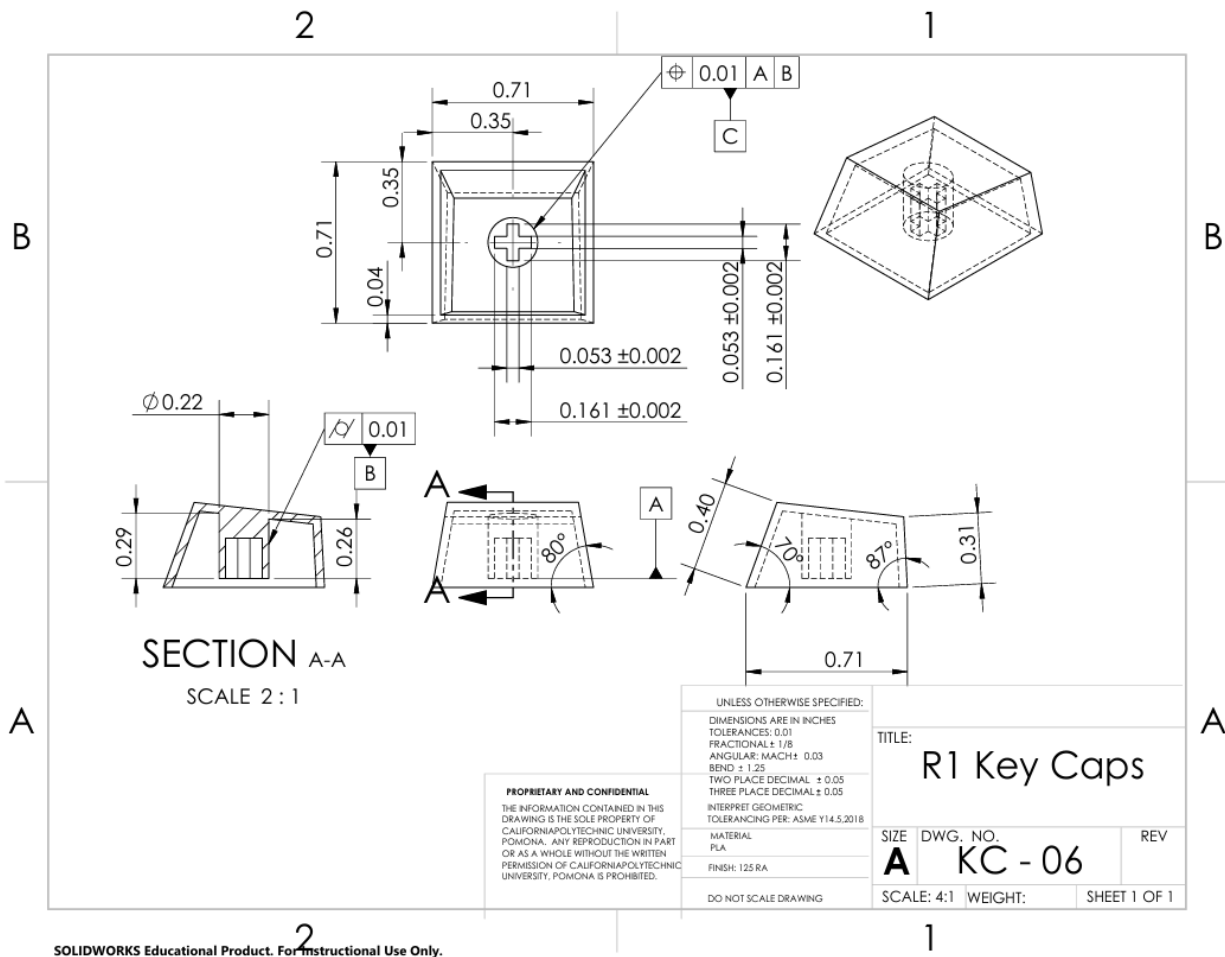
All part and Final assembly drawings

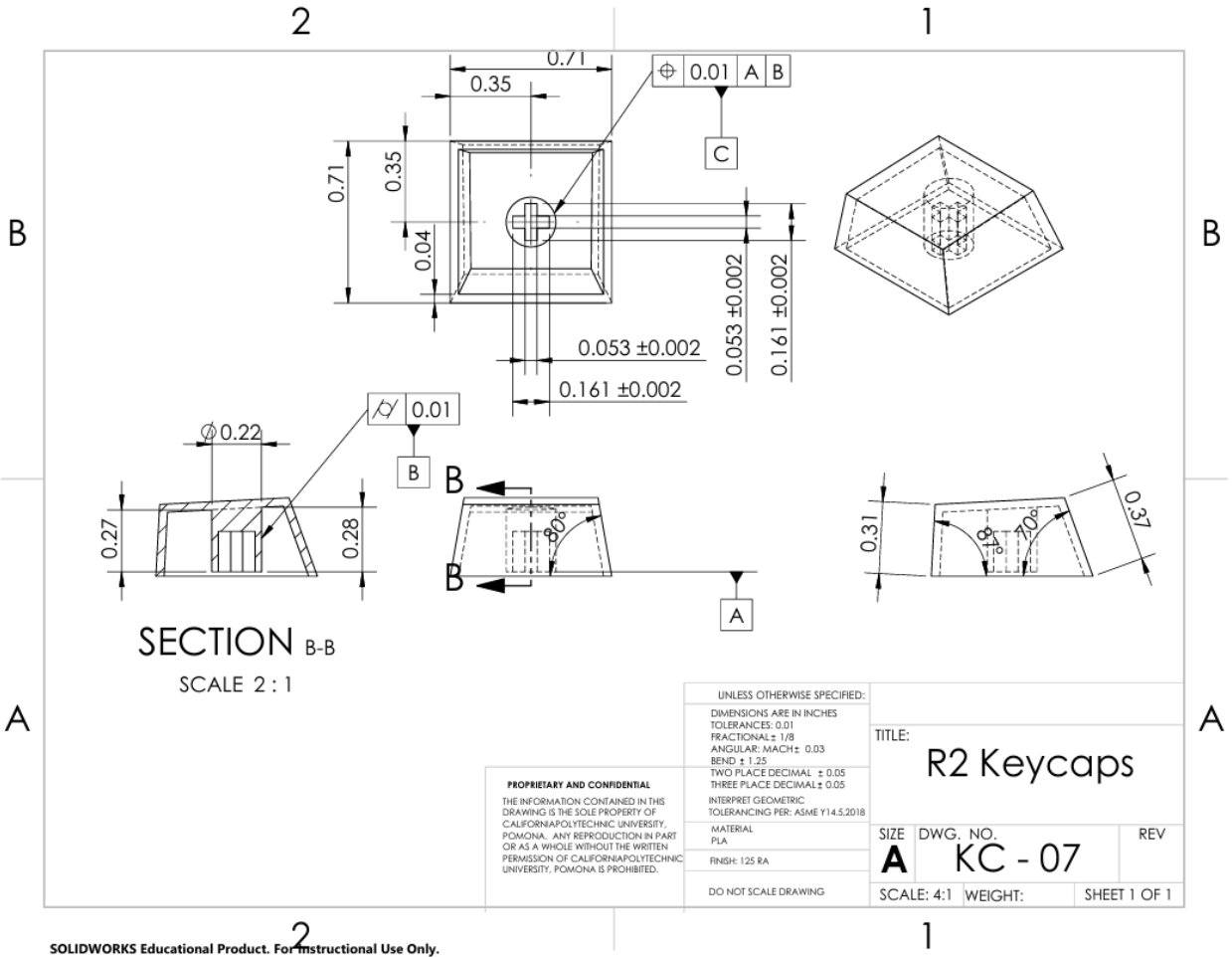


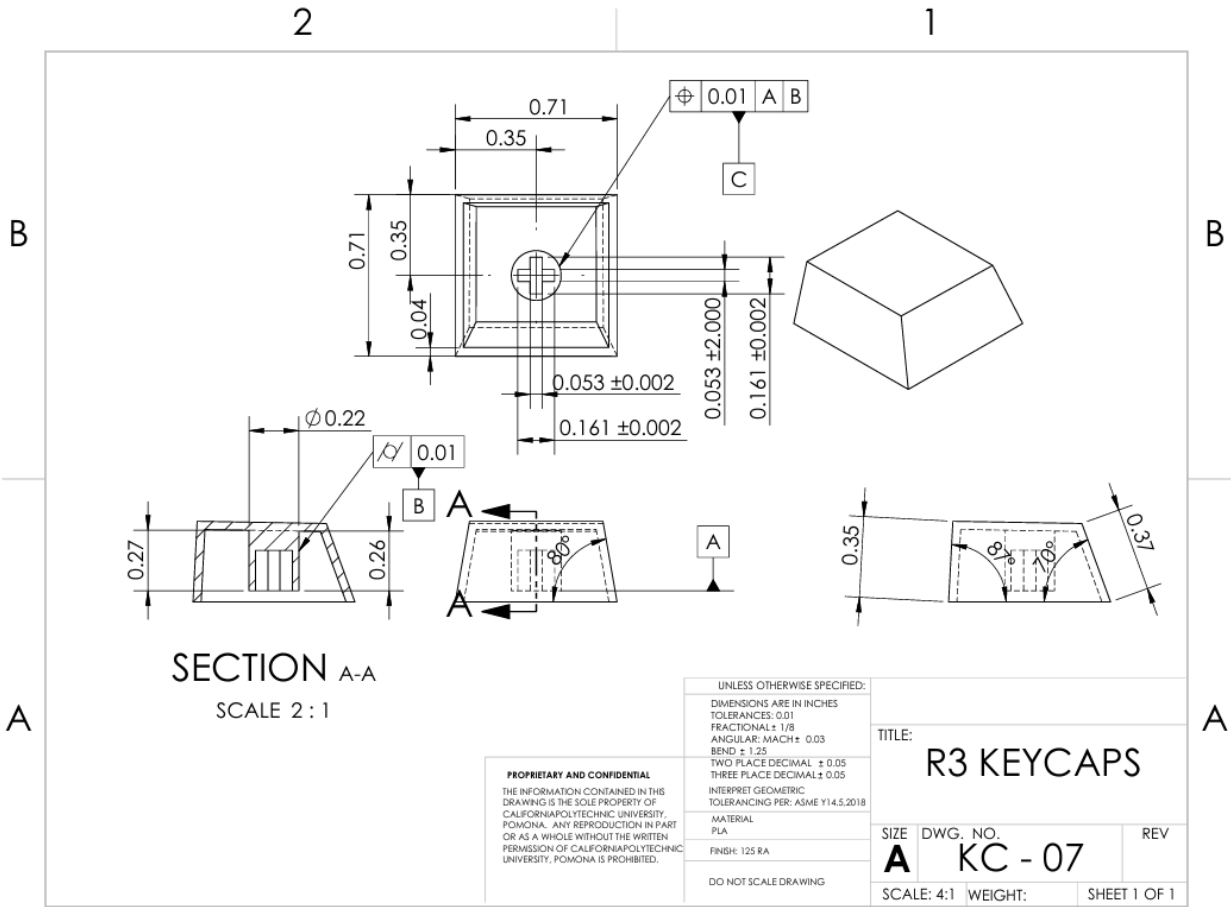


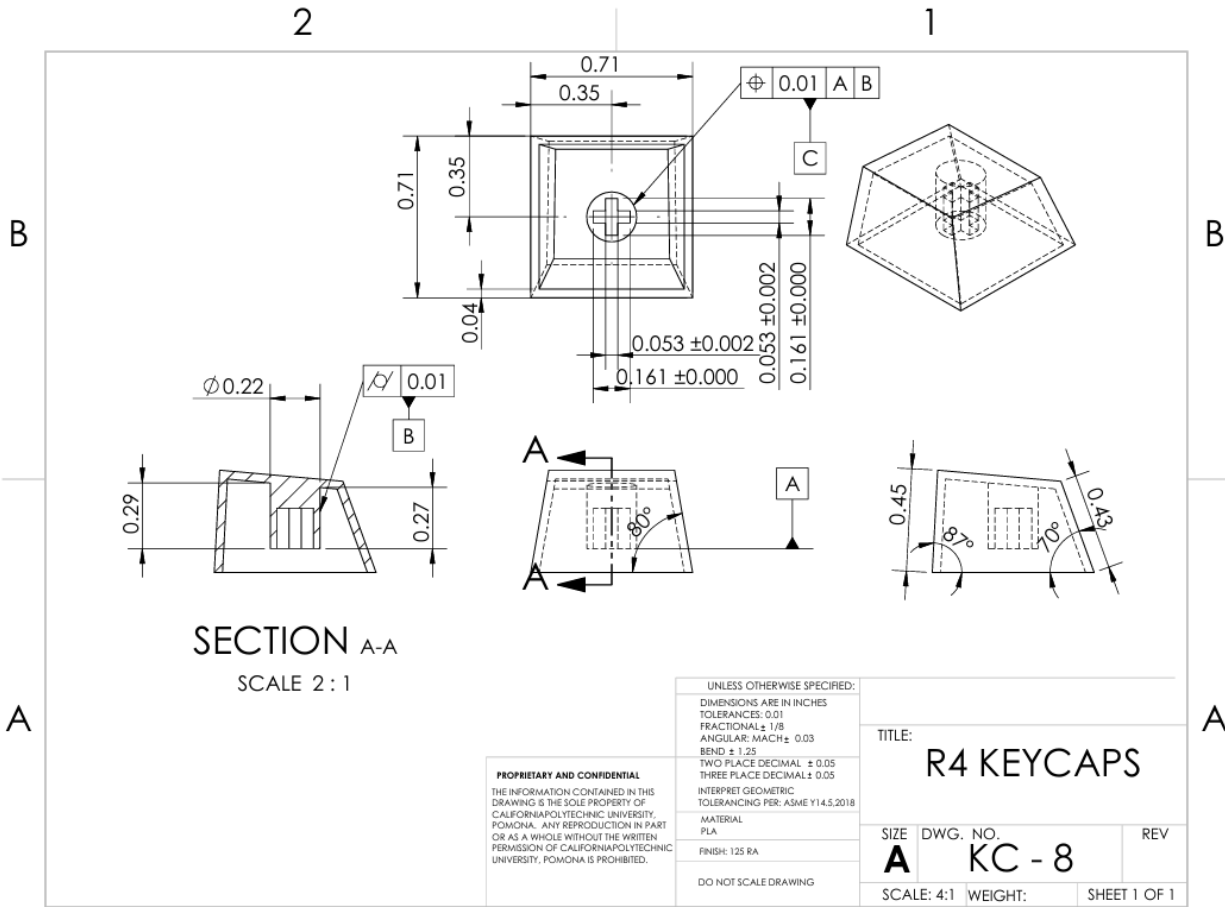


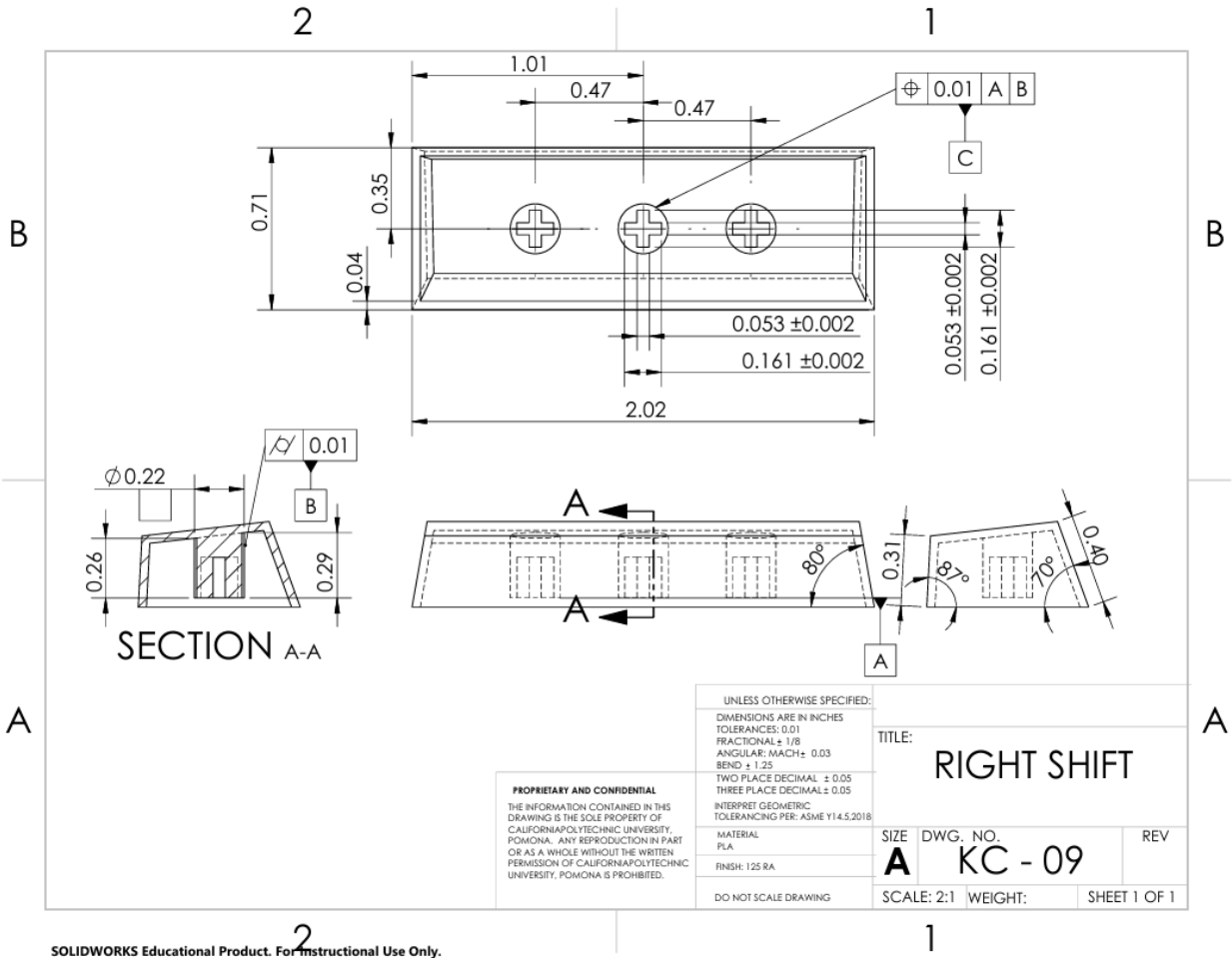


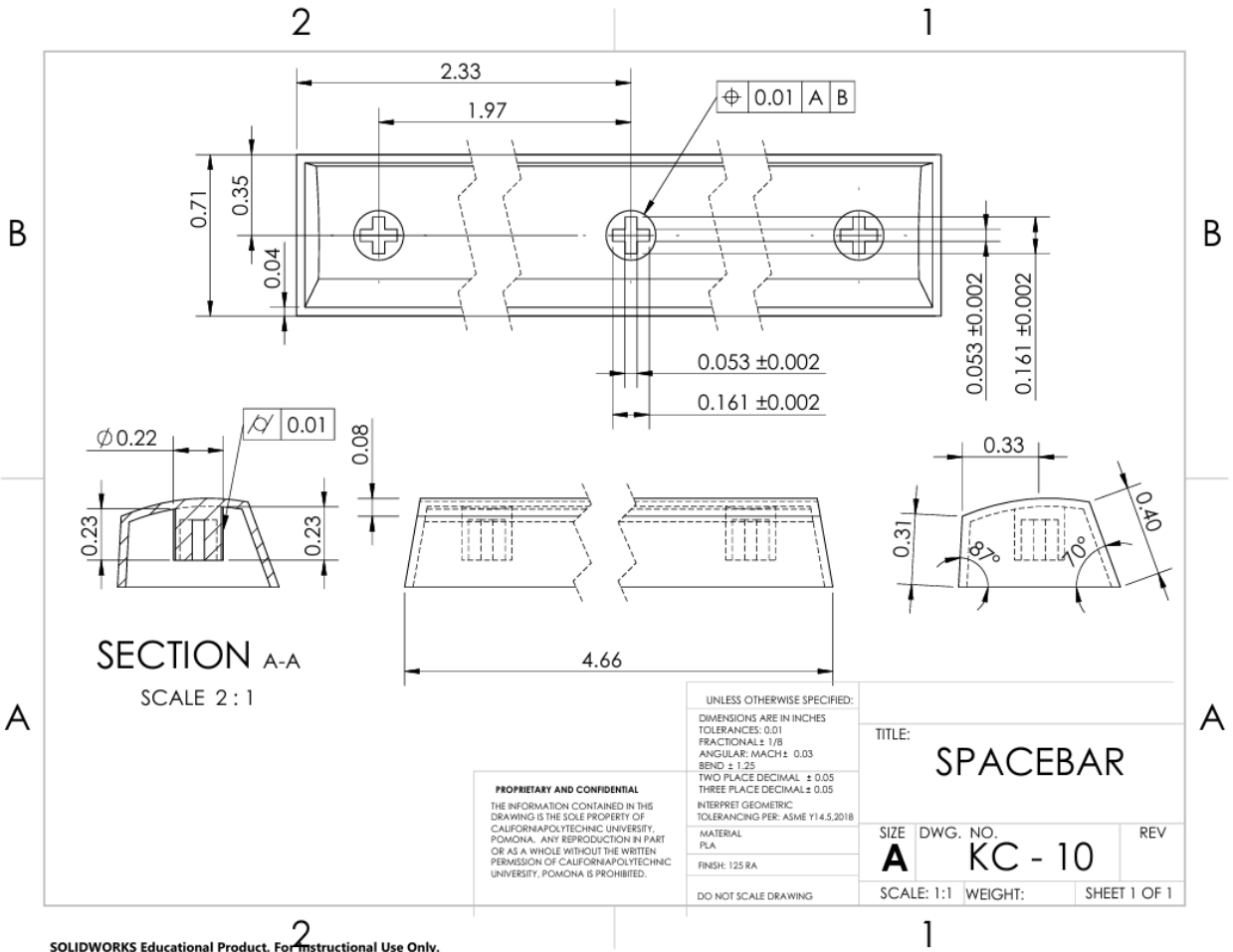


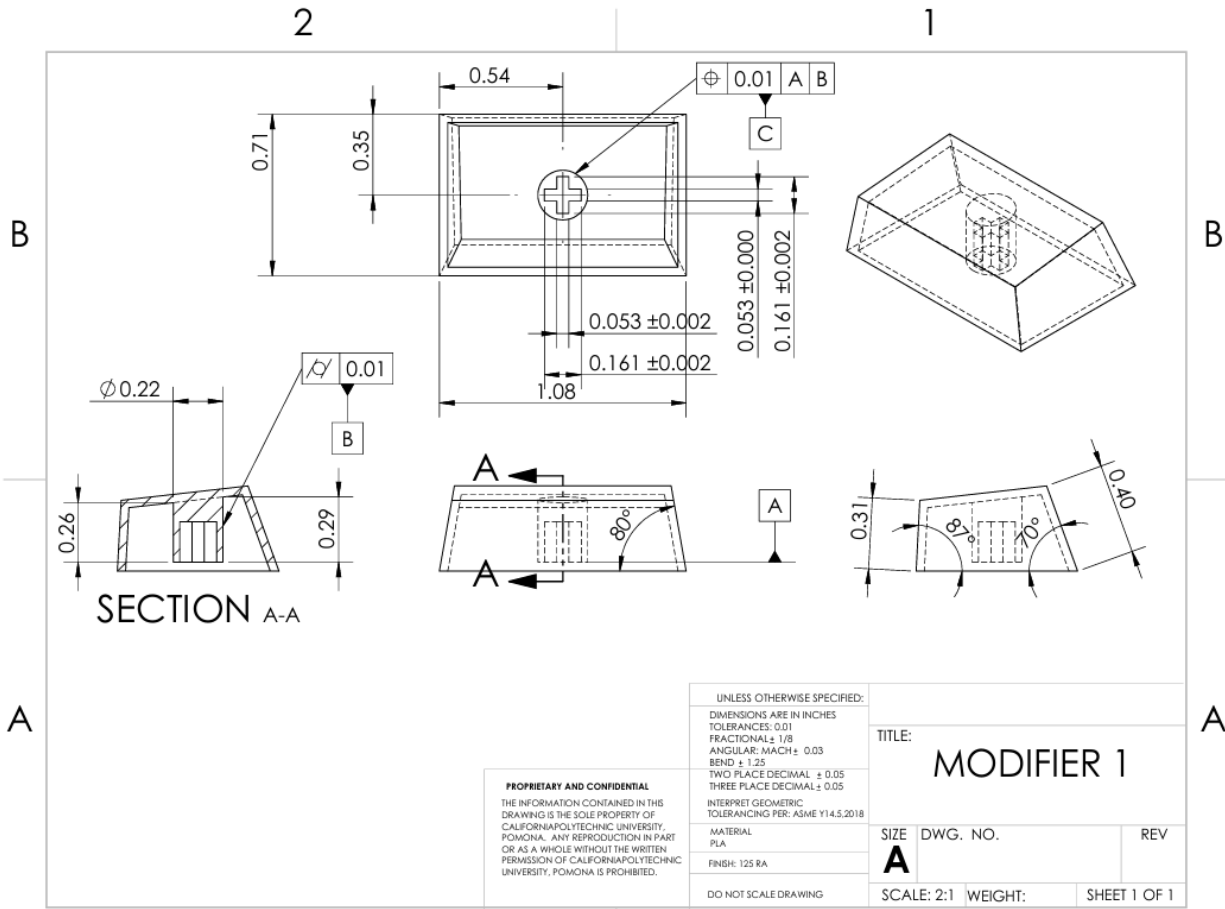


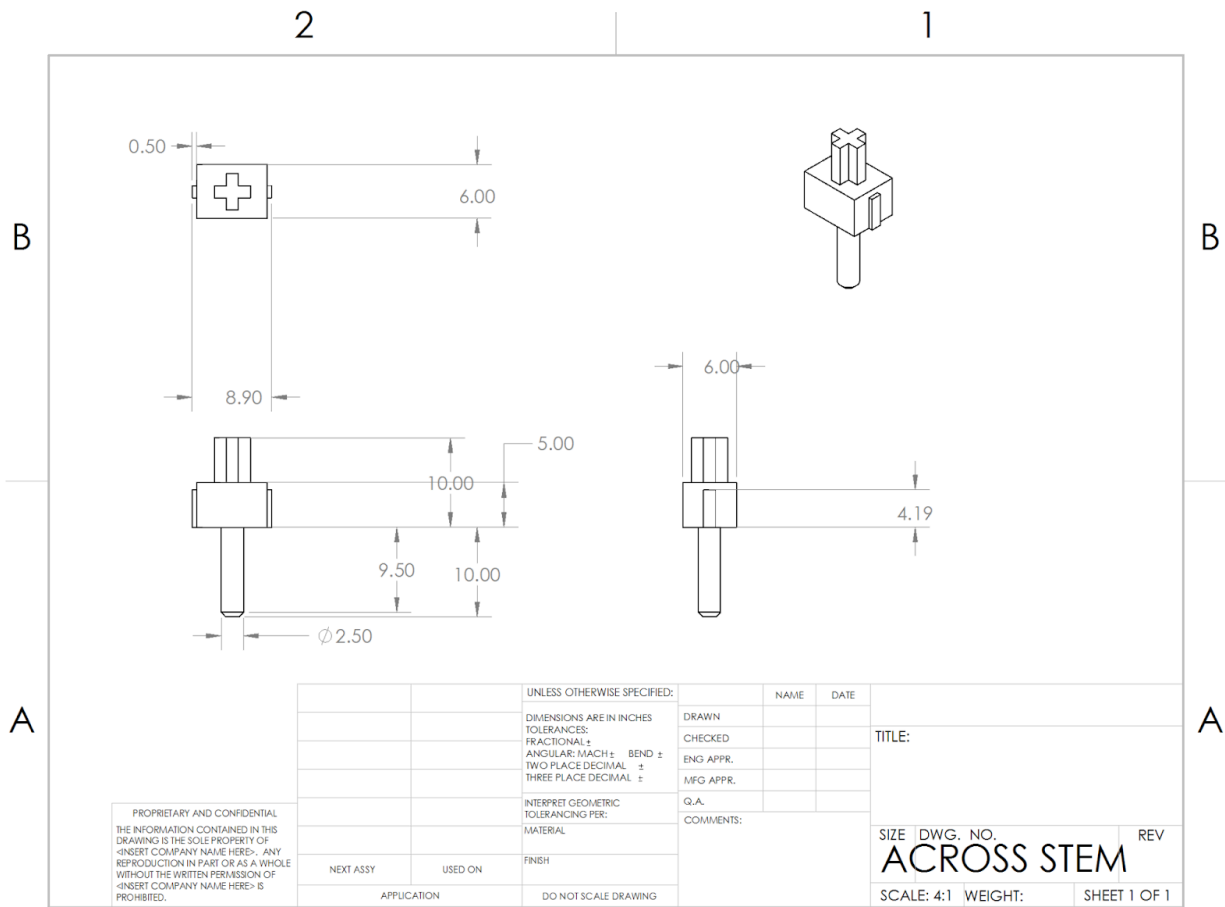












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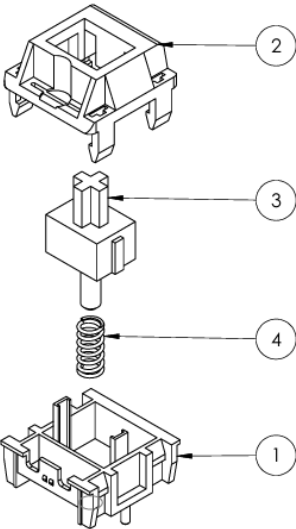
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B

B

A

A



ITEM NO.	PART NUMBER	DESCRIPTION	QTY.
1	SwitchBottom	Bottom Housing of Switch	1
2	Switch Top	Top Housing of Switch	1
3	CROSS STEM	Switch Stem	1
4	9001T11	Steel Precision Compression Spring	1

UNLESS OTHERWISE SPECIFIED:		NAME	DATE
DRAWN	QDLV		12/13/24
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ENG APPR.			
MFG APPR.			
Q.A.			
COMMENTS:			

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NEXT ASSY	USED ON

APPLICATION

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TITLE:

Switch Exploded Assembly

SIZE

DWG. NO.

REV

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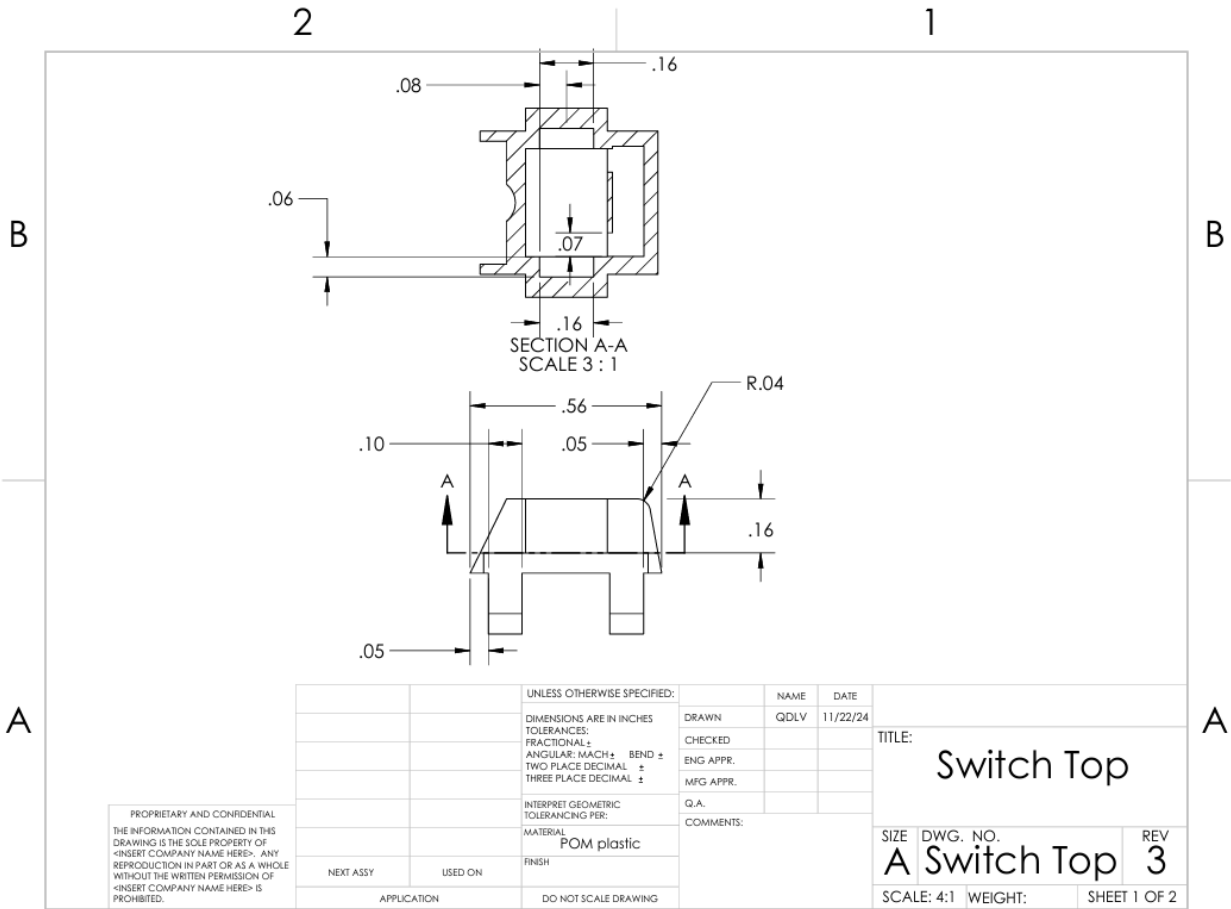
Switch

3

SCALE: 1:1

WEIGHT:

SHEET 2 OF 2



2

14.00

10.60

14.20

14.00

11.00

1

3

2

ITEM NO.	PART NUMBER	DESCRIPTION	QTY.
1	SwitchBottom	Bottom Housing of Switch	1
2	Switch Top	Top Housing of Switch	1
3	CROSS STEM	Switch Stem	1
4	9001T11	SteelPrecision Compression Spring	1

UNLESS OTHERWISE SPECIFIED:

DIMENSIONS ARE IN INCHES

TOLERANCES:

FRACTIONAL: ±

ANGULAR: MACH ± BEND ±

TWO PLACE DECIMAL: ±

THREE PLACE DECIMAL: ±

INTERPRET GEOMETRIC TOLERANCING PER:

MATERIAL:

FINISH:

DO NOT SCALE DRAWING

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ENG APPR.

MFG APPR.

Q.A.

COMMENTS:

NAME

QDLV

13/12/24

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NEXT ASSY

USED ON

APPLICATION

TITLE:

Switch Assembly

SIZE

DWG. NO.

REV

A

Switch

4

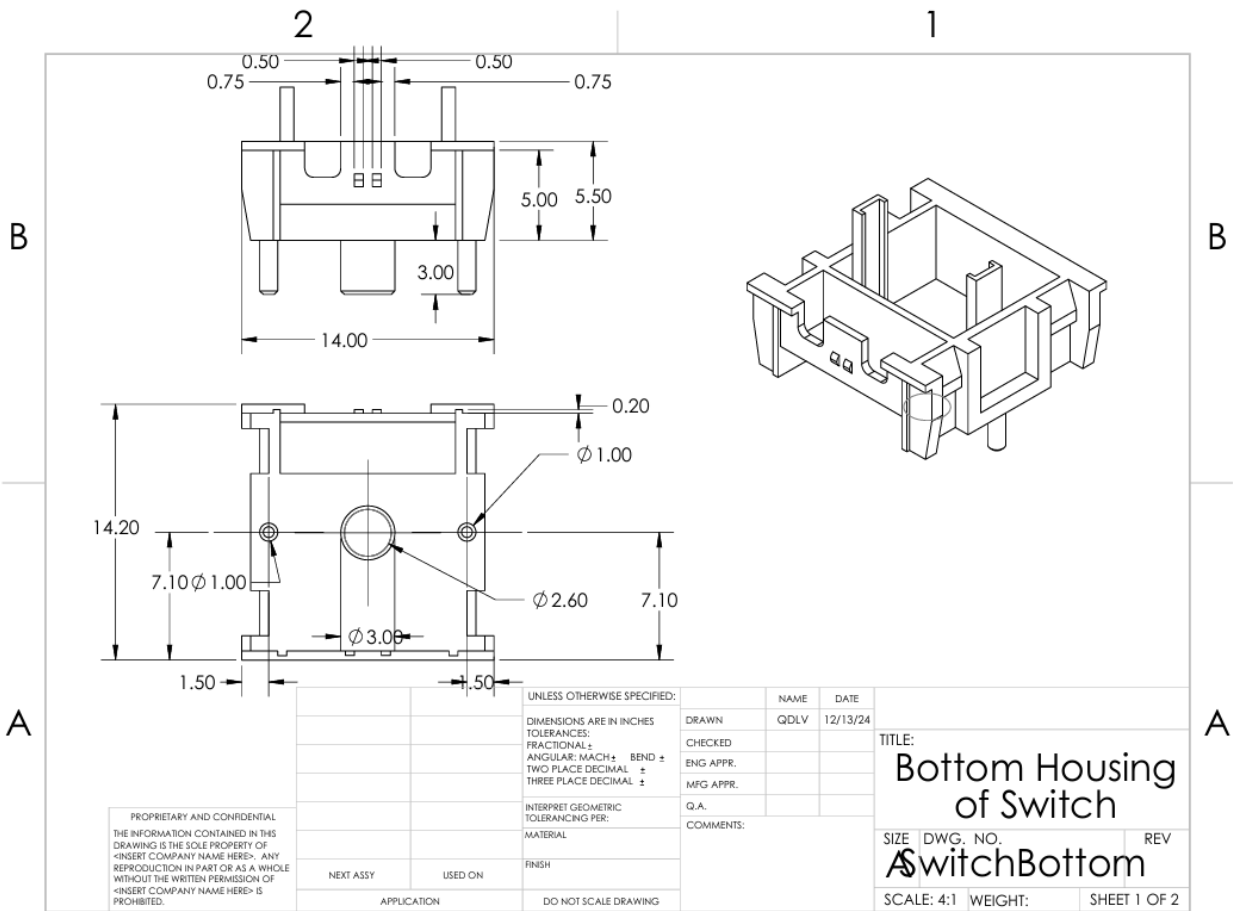
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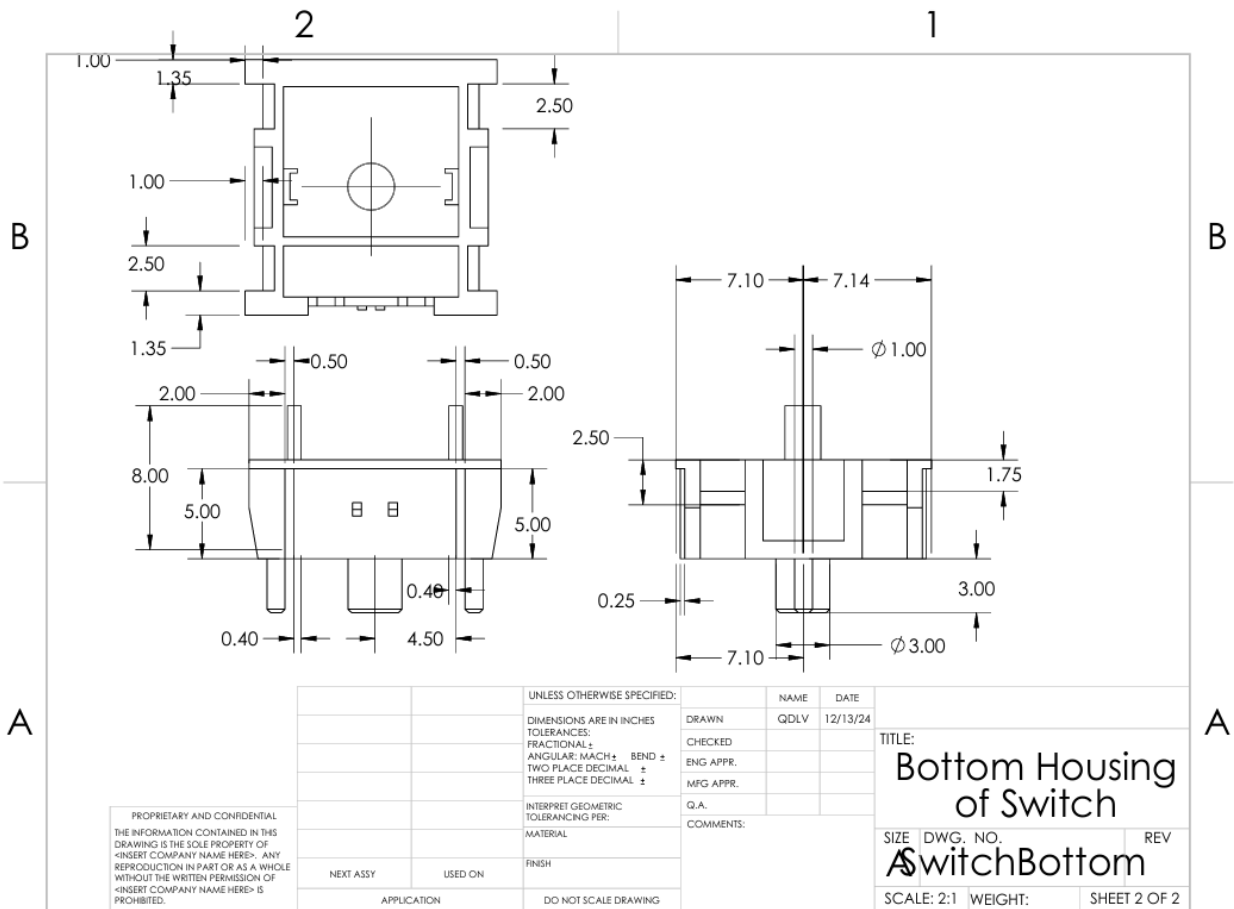
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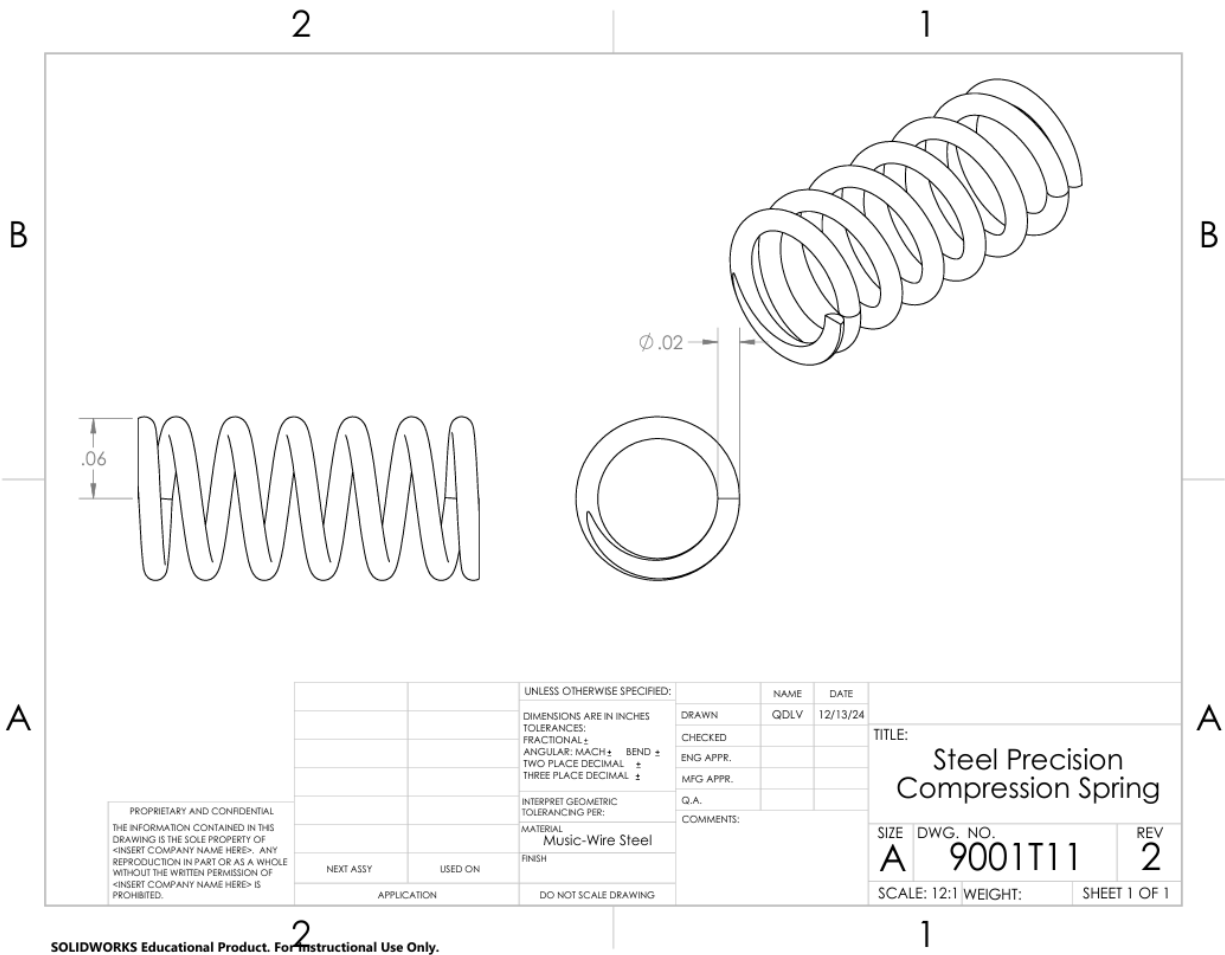
SHEET 1 OF 2

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Conclusions

In conclusion, the process of modeling the keyboard and its components proved to be an incredibly time-consuming and effort-intensive undertaking, far exceeding our initial expectations. From the intricate mechanisms of the key switch to the complexity of the PCB board, every step required an extraordinary level of precision, patience, and attention to detail. Reverse engineering the keyswitch alone was already challenging, involving numerous revisions to ensure all dimensions, tolerances, and internal mechanisms were accurately captured. However, tackling the PCB board added an entirely new level of difficulty. The meticulous documentation of component placements, and the need to ensure every hole was correct made

the process feel overwhelming at times. Even small errors required revisiting earlier stages of the design, further extending the time required to complete the task.

While this project did deepen our understanding of the complexities involved in reverse engineering and electronic design, it felt disproportionate to the scope of the class and the skills we were meant to develop. Focusing on just the keyswitch or a similarly simpler component would have been more aligned with the course objectives and allowed us to concentrate on mastering foundational skills without becoming bogged down by the exhaustive details of a full keyboard system. Such an approach would not only have been more practical but also far more effective in reinforcing the core principles of CAD modeling. The scope of this project ultimately highlighted the importance of balancing complexity with learning objectives to ensure time and effort are spent meaningfully in a way that enhances understanding without unnecessary frustration.